

EMCH 260

SOLID MECHANICS

Ming Hu

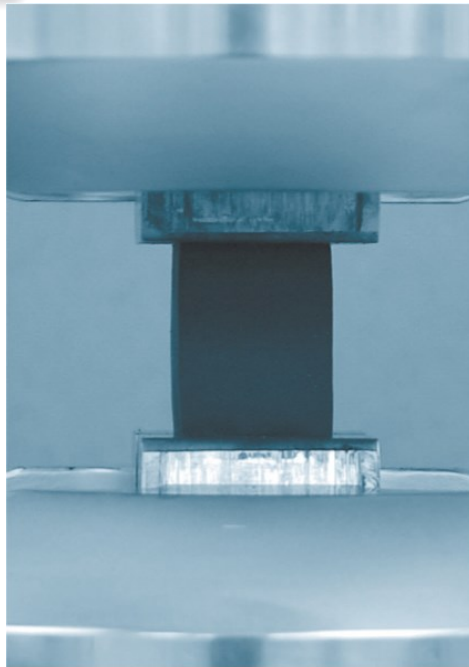
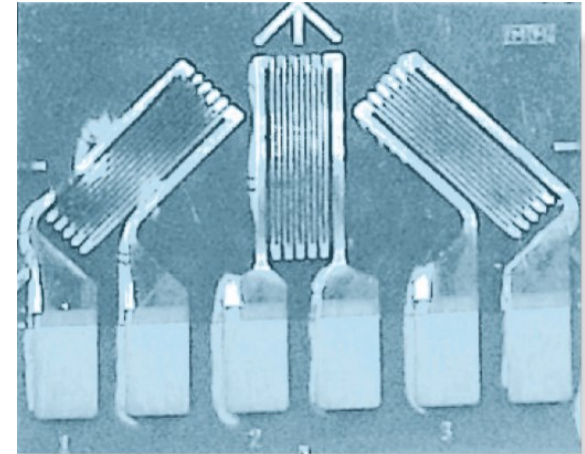
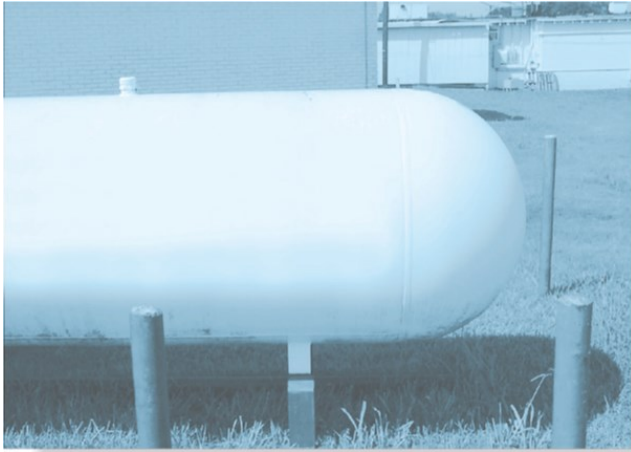
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Strain Transformation

Chapter 10.1 & 10.2

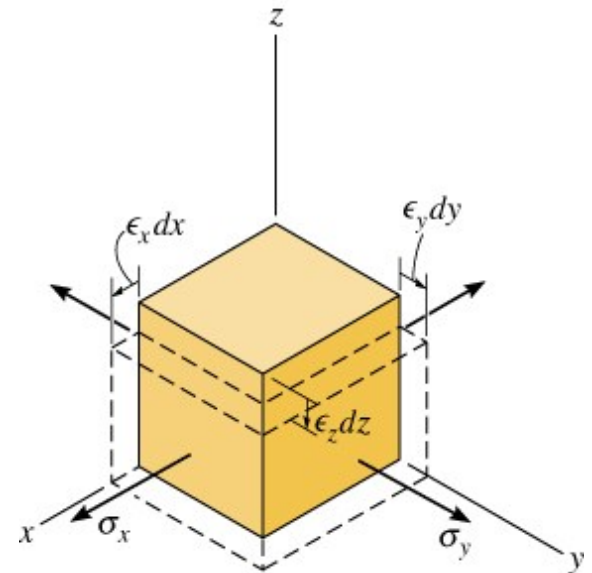
- ✓ Apply the stress transformation methods derived in Chapter 9 to similarly transform strain
- ✓ Determine principal strains and maximum in-plane shear strain
- ✓ Determine the absolute maximum shear strain in 2D and 3D cases

APPLICATIONS



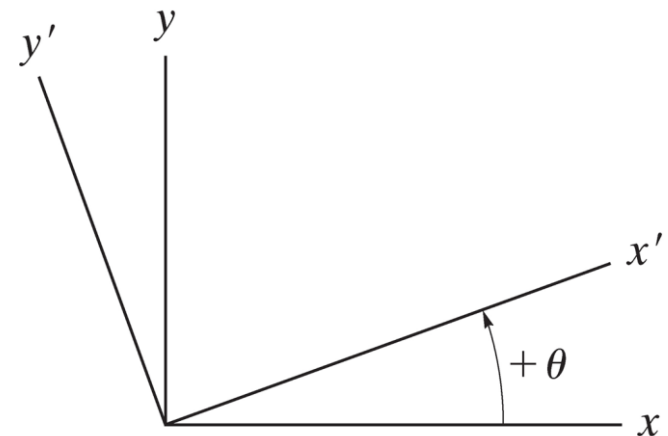
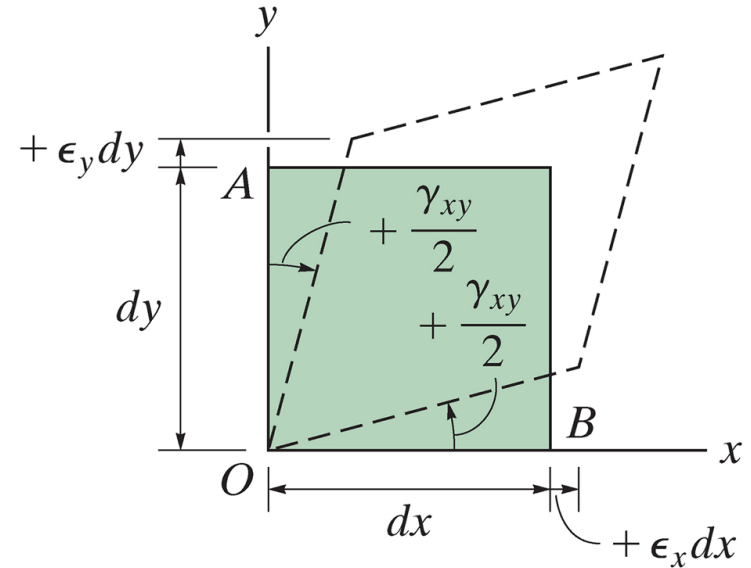
GENERAL STATE OF STRAIN AT A POINT

- General state of strain at a point is characterized by six independent normal and shear strain components; ϵ_x , ϵ_y , ϵ_z , γ_{xy} , γ_{yz} , and γ_{zx}
- A plane-strained element is subjected to two components of normal strain (ϵ_x , ϵ_y) and one component of shear strain, γ_{xy} .
- **Plane stress does not cause plane strain!**



SIGN CONVENTION

- To use the same convention as defined in textbook Chapter 2.2.
- With reference to differential element shown, normal strains ϵ_x and ϵ_y are positive if they cause **elongation** along the x and y axes.
- Shear strain γ_{xy} is positive if the interior angle AOB becomes **smaller** than 90° .
- Similar to plane stress, when measuring the normal and shear strains relative to the x' and y' axes, the angle θ will be positive provided it follows the curling of the right-hand fingers, **counterclockwise**.



GENERAL EQUATION OF PLANE-STRAIN TRANSFORMATION

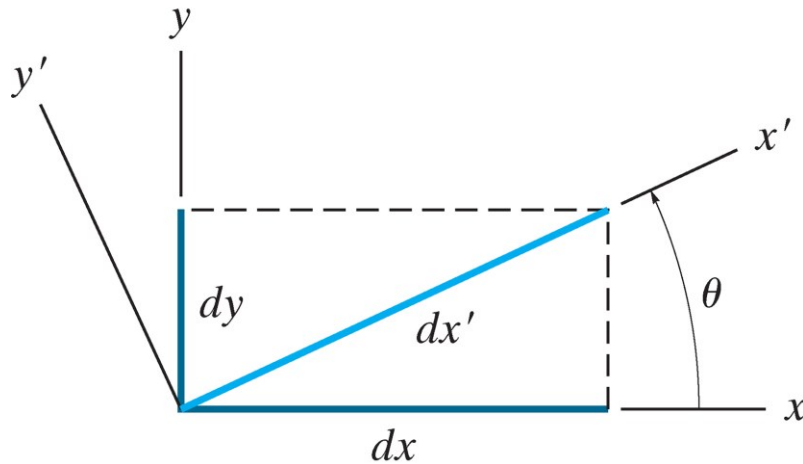
Normal and shear strains

- Components of line dx and dx' are elongated and we add all elongations together.

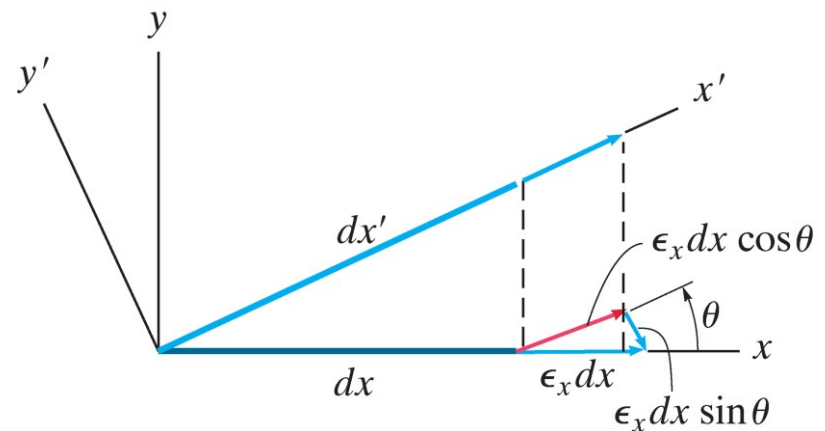
$$\delta x' = \varepsilon_x dx \cos \theta + \varepsilon_y dy \sin \theta + \gamma_{xy} dy \cos \theta$$

- The normal strain along the line dx' is $\varepsilon_{x'} = \delta x' / dx'$. Using Eqn 10-1,

$$\varepsilon_{x'} = \varepsilon_x \cos^2 \theta + \varepsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cos \theta \quad (10-2)$$



Before deformation



Normal strain ε_x

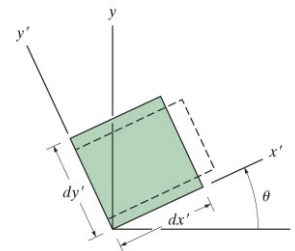
GENERAL EQUATION OF PLANE-STRAIN TRANSFORMATION, CONT'D

Normal and shear strains

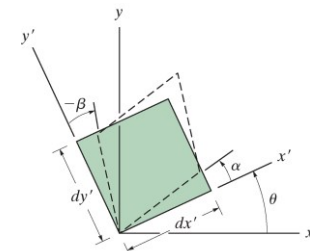
$$\epsilon_{x'} = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta \quad (10-5)$$

$$\frac{\gamma_{x'y'}}{2} = -\left(\frac{\epsilon_x - \epsilon_y}{2} \right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta \quad (10-6)$$

$$\epsilon_{y'} = \frac{\epsilon_x + \epsilon_y}{2} - \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta - \frac{\gamma_{xy}}{2} \sin 2\theta \quad (10-6)$$



Positive normal strain, $\epsilon_{x'}$



Positive shear strain, $\gamma_{x'y'}$

PRINCIPAL STRAIN

Principal strains

- We can orientate an element at a point such that the element's deformation is only represented by normal strains, with no shear strains.
- The material must be isotropic, and the axes along which the strains occur must coincide with the axes that define the principal axes.
- Thus from Eqs. 9-4 and 9-5,

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \quad (10-8)$$

(Orientation of principal planes)

PRINCIPAL STRAIN, CONT'D

Principal strains

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} \quad (10-9)$$

Maximum in-plane shear strain

- Using Eqns 9-6, 9-7 and 9-8, we get

$$\tan 2\theta_s = -\left(\frac{\epsilon_x - \epsilon_y}{\gamma_{xy}}\right) \quad (10-10)$$

$$\frac{\gamma_{\text{in-plane}}^{\text{max}}}{2} = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} \quad (10-11)$$

PRINCIPAL STRAIN, CONT'D

Average normal strain

- Using Eqns 9-6, 9-7 and 9-8, we get

$$\epsilon_{\text{avg}} = \frac{\epsilon_x + \epsilon_y}{2} \quad (10-12)$$

IMPORTANT NOTES

IMPORTANT

- Due to Poisson effect, the state of plane strain is not a state of plane stress, and vice versa.
- A point on a body is subjected to plane stress when the surface of the body is stress-free.
- Plane strain analysis may be used within the plane of the stresses to analyze the results from the gauges. Remember though, there is normal strain that is perpendicular to the gauges.
- When the state of strain is represented by the principal strains, no shear strain will act on the element.
- The state of strain at the point can also be represented in terms of the maximum in-plane shear strain. In this case, an average normal strain will also act on the element.
- The element representing the maximum in-plane shear strain and its associated average normal strains is 45° from the element representing the principal strains.

MOHR'S CIRCLE – PLANE STRAIN

$$\left[\epsilon_{x'} - \left(\frac{\epsilon_x + \epsilon_y}{2} \right) \right]^2 + \left(\frac{\gamma_{x'y'}}{2} \right)^2 = R^2$$

Analog between plane stress and plane strain

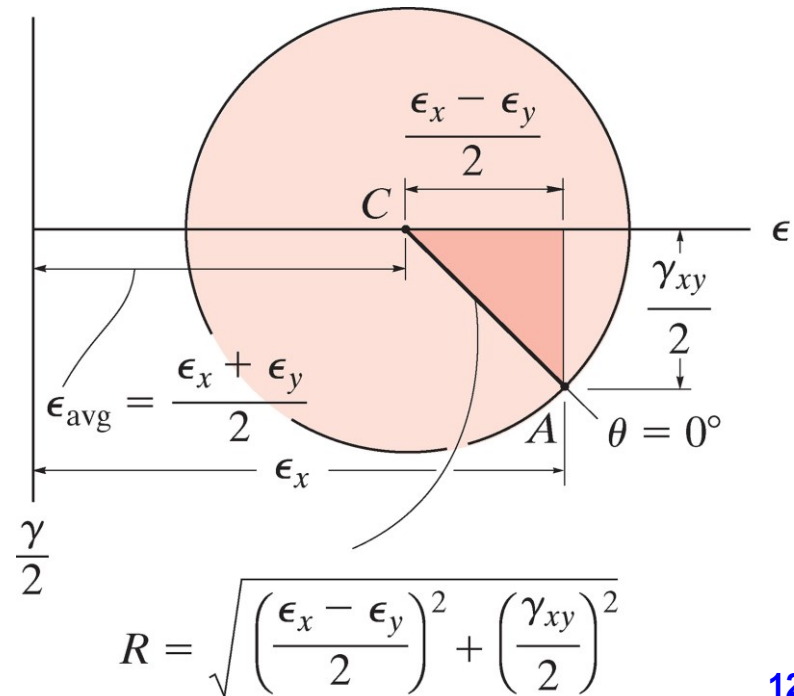
Since ϵ_x , ϵ_y and γ_{xy} are *known constants*, the above equation can be written in a more compact form as

$$(\epsilon_{x'} - \epsilon_{avg})^2 + \left(\frac{\gamma_{x'y'}}{2} \right)^2 = R^2$$

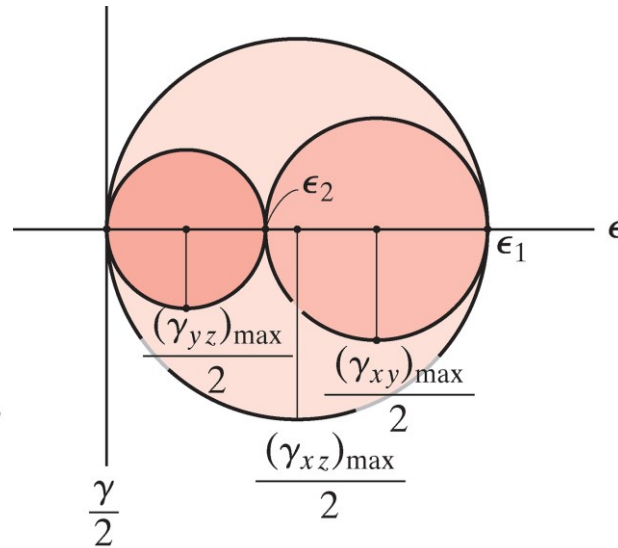
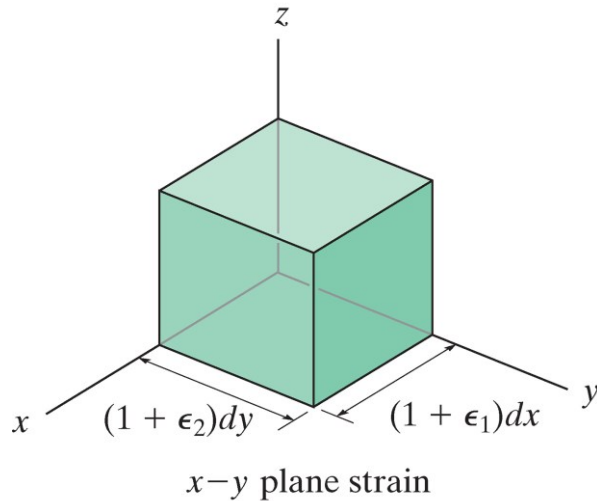
$$\epsilon_{avg} = \frac{\epsilon_x + \epsilon_y}{2}$$

$$R = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2} \right)^2 + \left(\frac{\gamma_{xy}}{2} \right)^2}$$

Plane Stress	Plane Strain
σ_x	ϵ_x
σ_y	ϵ_y
τ_{xy}	$\frac{\gamma_{xy}}{2}$

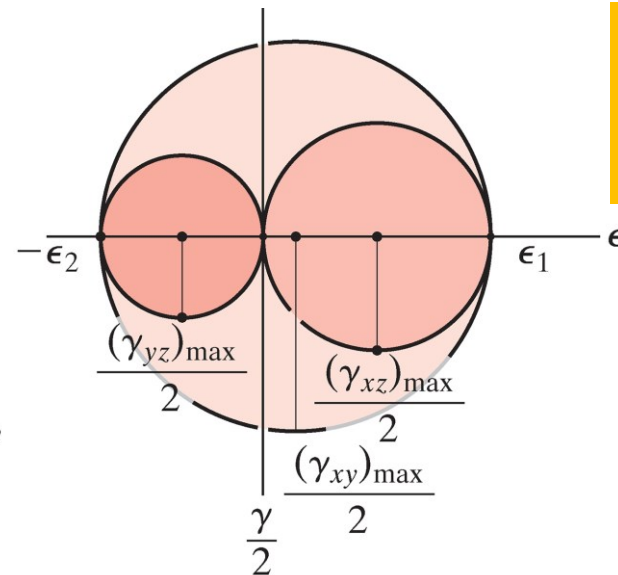
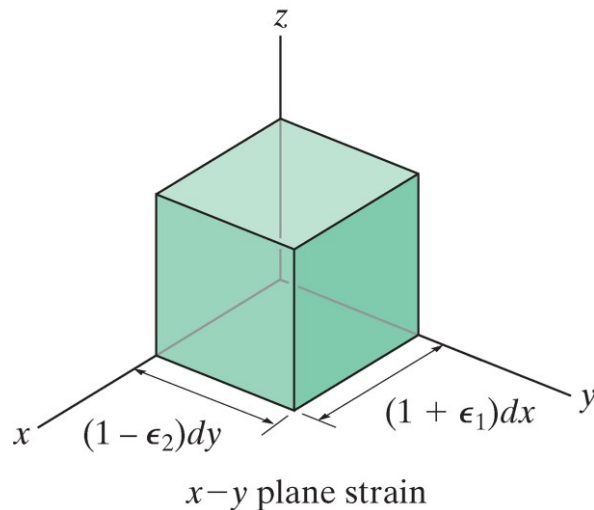


ABSOLUTE MAXIMUM SHEAR STRAIN



$$\gamma_{\max}^{\text{abs}} = (\gamma_{xz})_{\max} = \epsilon_1$$

ϵ_1 and ϵ_2 have
 same sign and
 $\epsilon_1 > \epsilon_2$



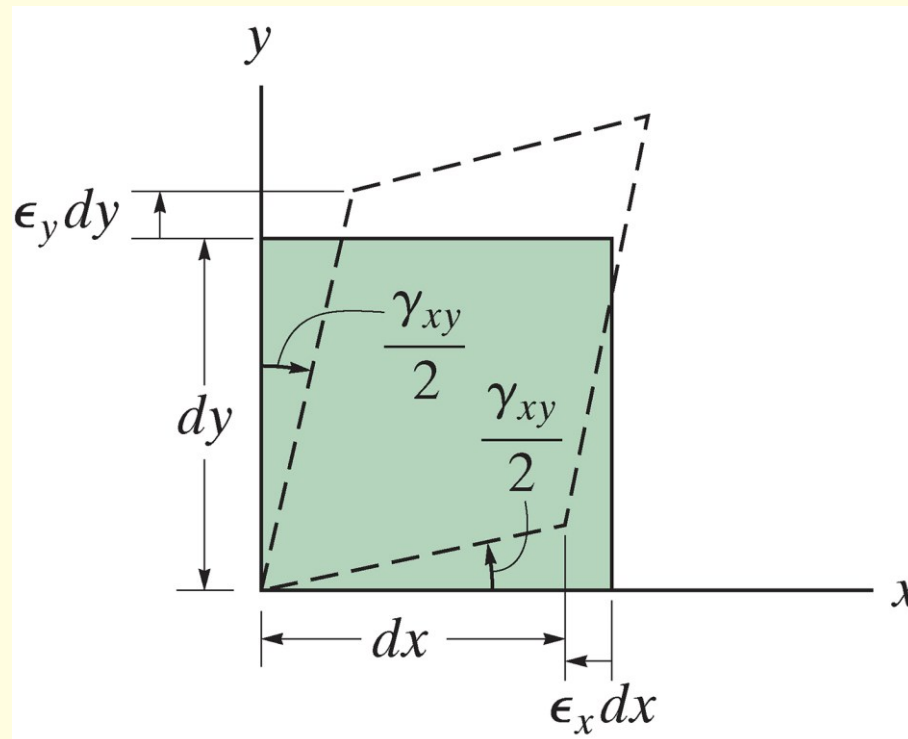
$$\gamma_{\max}^{\text{abs}} = (\gamma_{xy})_{\text{in-plane max}} = \epsilon_1 - \epsilon_2$$

ϵ_1 and ϵ_2 have
 opposite sign

IN-CLASS EXAMPLE

The state of plane strain at a point has strain components of $\epsilon_x = -200 \times 10^{-6}$, $\epsilon_y = 400 \times 10^{-6}$, and $\gamma_{xy} = 120 \times 10^{-6}$.

Determine the maximum in-plane shear strain and the corresponding orientation of the principal plane, and the absolute maximum shear strain at the point.



IN-CLASS EXAMPLE, cont'd

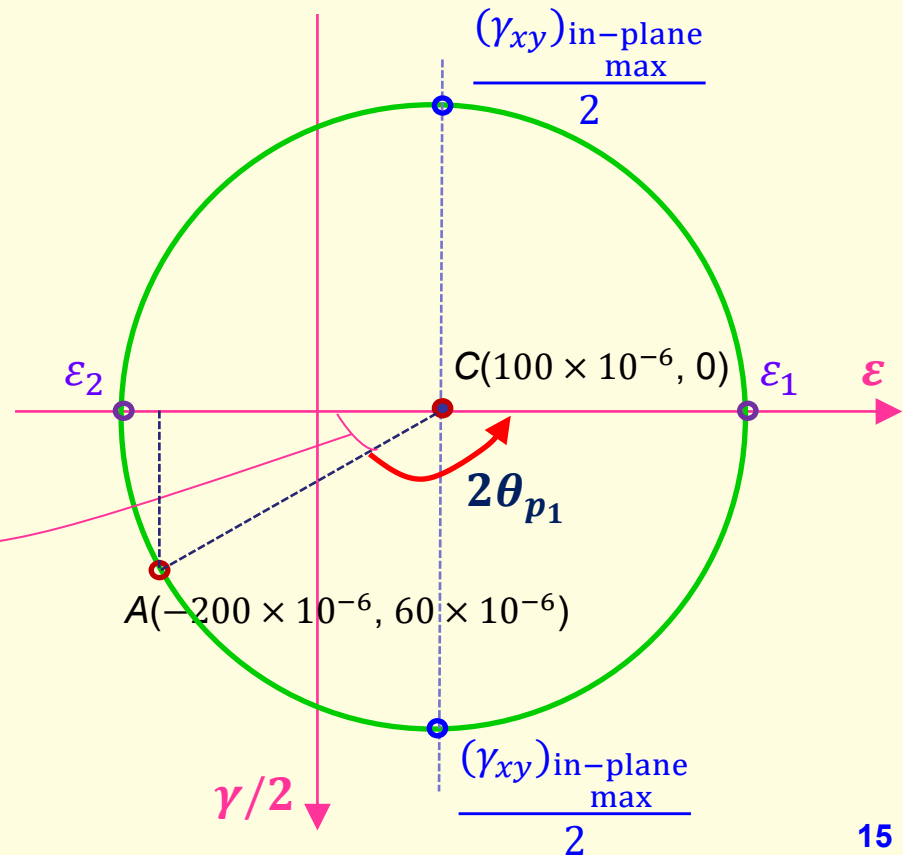
Solutions

Step 1: Construction of the Mohr's Circle of Plane Strain

Given $\varepsilon_x = -200 \times 10^{-6}$, $\varepsilon_y = 400 \times 10^{-6}$, $\gamma_{xy} = 120 \times 10^{-6}$, the center of the Mohr's circle C is located on the ε -axis, at the point $\varepsilon_{\text{avg}} = \frac{\varepsilon_x + \varepsilon_y}{2} = 100 \times 10^{-6}$

The point $C(100 \times 10^{-6}, 0)$ and the reference point $A(-200 \times 10^{-6}, 60 \times 10^{-6})$ are labeled. Then, the radius of the circle is $R = \sqrt{(100 + 200)^2 + 60^2} \times 10^{-6} = 306 \times 10^{-6}$, and the Mohr's circle is constructed (right figure).

$$\tan^{-1} 60/300 = 11.3^\circ$$



IN-CLASS EXAMPLE, cont'd

Solutions, Cont'd

Step 2: Principal Strains and Orientation of Principal Plane

$$\varepsilon_1 = (100 + 306) \times 10^{-6} = 406 \times 10^{-6} \quad \textbf{(Ans.)}$$

$$\varepsilon_2 = (100 - 306) \times 10^{-6} = -206 \times 10^{-6} \quad \textbf{(Ans.)}$$

The maximum in-plane shear stress is

$$\gamma_{\max}^{\text{in-plane}} = \varepsilon_1 - \varepsilon_2 = 612 \times 10^{-6} \quad \textbf{(Ans.)}$$

From the circle, the **counterclockwise** angle $2\theta_{p_1}$, measured from CA to the $+\varepsilon$ axis, is

$$2\theta_{p_1} = 180^\circ - \tan^{-1}\left(\frac{60}{300}\right) = 168.7^\circ$$

$$\text{Thus, } \theta_{p_1} = 84.35^\circ \quad \textbf{(Ans.)}$$

The counterclockwise rotation defines the direction of the **x' -axis and ε_1** .

IN-CLASS EXAMPLE, cont'd

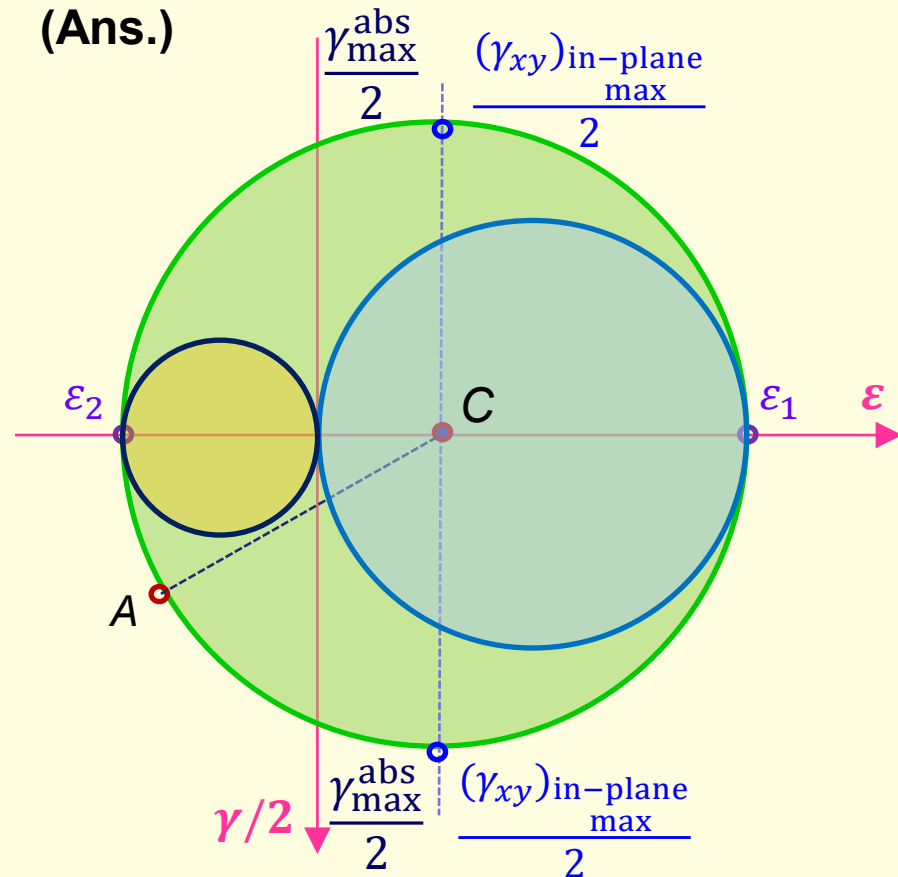
Solutions, Cont'd

Step 3: Absolute Maximum Shear Strain

Since ε_1 and ε_2 have opposite signs, the absolute maximum shear strain is:

$$\gamma_{\max}^{\text{abs}} = \varepsilon_1 - \varepsilon_2 = 612 \times 10^{-6}$$

(Ans.)



RELEVANT PRACTICE

- **Assignment 4 on Blackboard, problem #1-2 (go to Blackboard under the “Assignments” folder, **Assignment 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**
- **Optional Extra Exercise 4 on Blackboard, problem #1-2 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 493 – 511**
 - **In particular, Examples 10.1 – 10.7**